

Session 5 · Deck 5.A2

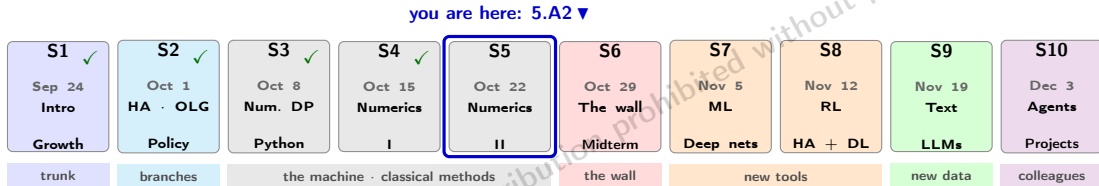
The Model, and the Parameter That Scales Risk

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Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are



Build it on paper → solve it on the machine → push it to the frontier with AI

Block A: 5.A1 what perturbation computes · **5.A2** the model and the parameter · 5.A3 first order · 5.A4 linear RE systems · 5.A5 higher order, risk, pruning · 5.A6 the worked example.

The Model We Will Expand

$$\begin{aligned} \max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \ln c_t \\ \text{s.t. } c_t + k_{t+1} = e^{z_t} k_t^\alpha \\ z_t = \rho z_{t-1} + \sigma \varepsilon_t, \quad \varepsilon_t \sim N(0, 1) \end{aligned}$$

Equilibrium conditions:

$$\begin{aligned} \frac{1}{c_t} &= \beta \mathbb{E}_t \left[\frac{1}{c_{t+1}} \alpha e^{z_{t+1}} k_{t+1}^{\alpha-1} \right] \\ c_t + k_{t+1} &= e^{z_t} k_t^\alpha \end{aligned}$$

States (k_t, z_t) . **Controls** (c_t, k_{t+1}) . The object we want is the pair of **policy functions** $c(k, z)$ and $k'(k, z)$.

Quant_Macro Lec. 9, "Stochastic neoclassical growth model". Calibration $\alpha = 0.36$, $\beta = 0.95$, as in S3.

Why this model

Log utility with **full depreciation** is the Brock–Mirman benchmark of 1.B1 — the one model in this course whose policy is known in closed form. Every method gets tested here first.

$\delta = 1$ is not realistic. It is diagnostic. 5.A6 moves to the quarterly calibration with $\delta = 0.025$, where no closed form exists.

The One Thing We Know Exactly

Guess and verify gives the exact policy:

$$c_t = (1 - \alpha\beta) e^{z_t} k_t^\alpha$$
$$k_{t+1} = \alpha\beta e^{z_t} k_t^\alpha$$

and therefore the deterministic steady state

$$k^* = (\alpha\beta)^{\frac{1}{1-\alpha}} = 0.1870$$
$$c^* = (\alpha\beta)^{\frac{\alpha}{1-\alpha}} - (\alpha\beta)^{\frac{1}{1-\alpha}}$$
$$z^* = 0.$$

The steady state is the point we expand around; the closed-form policy is the answer we will grade against.

Quant_Macro Lec. 9, "Solution and steady state". Ledger brock_mirman: $k^* = 0.187032$.

Two different objects. Almost every model has a computable steady state. Almost none has a closed-form *policy*. Perturbation needs only the first — which is why it generalizes.

1.B1: "If a method cannot recover $k' = \alpha\beta z k^\alpha$, do not trust it on anything harder." VFI was tested here in S3. Perturbation is tested here in three slides.

The Goal, Written as Functional Equations

Substituting the unknown policies into the equilibrium conditions gives a system in which the unknowns are *functions*:

$$\frac{1}{c(k_t, z_t)} = \beta \mathbb{E}_t \left[\frac{\alpha e^{\rho z_t + \sigma \varepsilon_{t+1}} k(k_t, z_t)^{\alpha-1}}{c(k(k_t, z_t), \rho z_t + \sigma \varepsilon_{t+1})} \right]$$

$$c(k_t, z_t) + k(k_t, z_t) = e^{z_t} k_t^\alpha$$

Read the composition

c appears *evaluated at next period's state*, which is itself the unknown $k(\cdot)$. That nesting is why this is a functional equation and not an algebraic one.

Once $c(\cdot)$ and $k(\cdot)$ are known, everything follows: simulate, compute moments, produce impulse responses, do welfare.

A Parameter That Scales the Risk

There is a problem with expanding in (k, z) alone: we want to know what **uncertainty itself** does, and uncertainty is not a state.

The device is to introduce an artificial parameter χ multiplying the shock:

$$z_t = \rho z_{t-1} + \chi \sigma \varepsilon_t$$

- $\chi = 1$: the model we care about.
- $\chi = 0$: the **deterministic** economy, where we know how to compute everything.

The policies become $c(k_t, z_t; \chi)$ and $k(k_t, z_t; \chi)$, and χ is treated as a third argument to differentiate with respect to.

Quant_Macro Lec. 9, "A perturbation solution".

Notation

The literature, including Fernández-Villaverde & Guerrón-Quintana, writes λ for this parameter. This course reserves λ for the **invariant distribution** (2.A3), so we write χ . Nothing else changes.

The move

Risk has been turned into a *parameter we can differentiate*. That is the entire trick, and everything in 5.A5 is a consequence of it.

Why Expand Around $\chi = 0$?

Because $\chi = 0$ is the one point where the answer is available: no uncertainty, a deterministic steady state, and policies we can compute directly.

Expanding around it measures **how a small amount of risk changes behaviour relative to the deterministic economy** — which is exactly the economic question.

The by-products are immediate:

- first order gives a **linear state-space system**, so impulse responses are one matrix multiplication;
- second-order moments follow in closed form, with no simulation.

And the price

Accuracy deteriorates with large shocks, strong nonlinearity, and distance from the steady state. We do not have to take that on trust — 5.A6 measures exactly where it happens.

Note what is being assumed: that the solution is *smooth in χ at $\chi = 0$* . Models with occasionally binding constraints are not.

What Exactly Gets Differentiated

We build a local approximation around $(k^*, 0; 0)$. Taking the equilibrium conditions

$$\frac{1}{c(k_t, z_t; \chi)} = \beta \mathbb{E}_t[\dots]$$
$$c(k_t, z_t; \chi) + k(k_t, z_t; \chi) = e^{z_t} k_t^\alpha$$

we differentiate with respect to k_t , z_t and χ , and evaluate at $(k^*, 0; 0)$.

The unknowns become coefficients. c_k , c_z , c_χ , k_k , k_z , k_χ are numbers, not functions — the partial derivatives of the true policies at one point.

Taylor's theorem, twice over

Once to justify writing the policy as a series at all, and once to say what the truncation error is. Both need the smoothness the IFT gave us in 5.A1.

“Solve the model” has become “compute a few derivatives at a point we already know”.

Quant_Macro Lec. 9, “Taylor's theorem” — including the two frames of that title carried in the predecessor deck and dropped from the current one.

The Expansion, Written Out

To second order, around $(k^*, 0; 0)$, writing $\hat{k}_t = k_t - k^*$:

$$\begin{aligned}c_t &= c^* + c_k \hat{k}_t + c_z z_t + c_\chi \chi \\&\quad + \frac{1}{2} \left(c_{kk} \hat{k}_t^2 + 2c_{kz} \hat{k}_t z_t + c_{zz} z_t^2 + 2c_{k\chi} \hat{k}_t \chi + 2c_{z\chi} z_t \chi + c_{\chi\chi} \chi^2 \right) + \dots \\k_{t+1} &= k^* + k_k \hat{k}_t + k_z z_t + k_\chi \chi \\&\quad + \frac{1}{2} \left(k_{kk} \hat{k}_t^2 + 2k_{kz} \hat{k}_t z_t + k_{zz} z_t^2 + 2k_{k\chi} \hat{k}_t \chi + 2k_{z\chi} z_t \chi + k_{\chi\chi} \chi^2 \right) + \dots\end{aligned}$$

First order

Keep the linear terms only: six unknowns, $\{c_k, c_z, c_\chi, k_k, k_z, k_\chi\}$. 5.A3 shows that two of them are zero.

$c_{\chi\chi}$ is the term that will carry **precautionary behaviour**. It is a second-order object, which is already the answer to “why go beyond first order”.

Measured: On This Model, the Linear Solution Is Not an Approximation

Take logs of the exact policy:

$$\ln k_{t+1} = \ln(\alpha\beta) + z_t + \alpha \ln k_t,$$

and since $\ln k^* = \ln(\alpha\beta)/(1 - \alpha)$,

$$\widehat{\ln k_{t+1}} = \alpha \widehat{\ln k_t} + z_t.$$

The true policy is exactly linear in logs. So a first-order perturbation in logs should return α and 1 — not approximately.

Solved numerically, it does:

Measured

k on $\hat{k}$, k on z	0.36, 1.00
c on $\hat{k}$, c on z	0.36, 1.00
deviation from exact	1.1×10^{-16}
policy error, <i>in levels</i>	4.4×10^{-16}
every second-order term	$< 10^{-9}$

Machine precision, over $k \in [0.25k^*, 2.5k^*]$ — a tenfold range, not a neighbourhood.

Ledger brock_mirman. Coefficients by generalized Schur; the second-order terms come from an independent residual-matching solve and vanish to the precision of its finite differences.

What That Benchmark Can and Cannot Tell You

What it certifies. The algebra, the code, the steady state, the sign conventions, the transposes. If perturbation cannot return 0.36 and 1.00 here, the implementation is wrong and nothing else matters.

What it cannot certify. Anything about accuracy. The error is zero *because the model is log-linear*, not because perturbation is good. A benchmark that a method solves exactly measures nothing about the method.

This is the same trap as 3.A2's beautifully converged value function with a bad policy: the diagnostic that looks most reassuring is the one carrying least information.

Ledger brock_mirman, field note.

Why $\delta = 1$ does this

Full depreciation plus log utility makes the saving *rate* constant at $\alpha\beta$. Nothing is left for curvature to act on, so certainty equivalence is exact and every higher-order term is zero.

Hence 5.A6 runs the quarterly model with $\delta = 0.025$, where the second-order terms are not zero and the error is not zero either.

Takeaways

- The model is the **Brock–Mirman benchmark** of 1.B1: log utility, $\delta = 1$, closed-form policy, steady state $k^* = 0.1870$.
- Perturbation needs only the **steady state**, not the closed form — which is exactly why it survives into models that have no closed form.
- Risk becomes differentiable by introducing χ with $z' = \rho z + \chi \sigma \varepsilon'$: $\chi = 0$ is the deterministic economy, $\chi = 1$ the one we want.
- “Solve the model” has become “find $\{c_k, c_z, c_\chi, k_k, k_z, k_\chi\}$, then the second-order terms, then...”.
- Measured: on this model the first-order solution reproduces the exact policy to 10^{-16} **over a tenfold range of capital**, and every second-order term is zero.
- Which is a **correctness** test, not an accuracy test. A benchmark solved exactly tells you nothing about how the method behaves where it is actually needed.

Bridge: Where the Equations Come From

We now have unknowns — six of them at first order — and no equations for them yet. The next deck writes the equilibrium conditions as a single object $F(k, z; \chi) = 0$, observes that it is an **identity**, and differentiates it.

Two results fall out that are worth waiting for: two of the six unknowns turn out to be **exactly zero**, and the remaining system is exactly square.

Next (5.A3)

$F \equiv 0 \Rightarrow F_k = F_z = F_\chi = 0$; six unknowns, four equations, and why that is not a mistake. Certainty equivalence, and what it costs.

Keep the closed form in view. Everything 5.A3 derives can be checked against α and 1.